Foundations of Generative AI (ICT4403) - Assignment 2

Autoregressive Word-Level Text Generation: Comparative Study of Simple RNN, Bidirectional RNN, LSTM, and GRU

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Dataset & Source:	The Adventures of Sherlock Holmes (Project Gutenberg EBook #1661) https://www.gutenberg.org/cache/epub/1661/pg1661.txt
GitHub Repository Link:	https://github.com/NiftyRunner/fgai-autoregressive-textgen

1. Executive Summary & Objective

This project implements, evaluates, and compares four foundational recurrent neural network architectures (Simple RNN, Bidirectional RNN, LSTM, and GRU) for autoregressive word-level next-word text generation in PyTorch. The objective is to analyze the trade-offs between architectural complexity, gating mechanisms, parameter counts, sequence modeling fidelity, training dynamics, and autoregressive text generation quality under strictly comparable experimental conditions.

Trained on Arthur Conan Doyle's 'The Adventures of Sherlock Holmes' (Project Gutenberg #1661) on an NVIDIA RTX 4060 GPU, the models were benchmarked across cross-entropy loss, perplexity, top-1 accuracy, top-5 accuracy, parameter count, GPU runtime, and generation quality at temperatures T = 0.5, 1.0, and 1.5 across 4 standardized prompts.

2. Corpus Analysis & Preprocessing Pipeline

The corpus was downloaded directly from Project Gutenberg (https://www.gutenberg.org/cache/epub/1661/pg1661.txt). All legal headers and footers were programmatically stripped using verified Gutenberg bounding markers, yielding 574,123 clean characters.

Text Preprocessing Highlights:

- Tokenization: Lowercase text tokenized into words and specific punctuation (. , ; : ! ?) as separate tokens, preserving contractions.
- Corpus Dimensions: 121,193 total tokens, 7,891 unique tokens, and 7,291 sentences (mean length: 15.6 tokens).
- Vocabulary: Capped at 7,893 entries including special tokens (<pad> = 0, <unk> = 1).
- Sliding Windows: Fixed sequence length SEQ_LEN = 25 tokens with the 26th token as target.

• Split Strategy: 80% train (96,929 windows), 10% validation (12,094 windows), 10% test (12,095 windows) partitioned sequentially to avoid data leakage.

3. Model Architectures & Standardized Hyperparameters

All four architectures adhere to an identical structure: nn.Embedding -> Recurrent Layer -> nn.Dropout(0.3) -> nn.Linear -> Logits. The last timestep output is passed to the classifier head for next-word prediction.

Model	Embedding	Hidden Size	Layers	Dropout	Parameters
SimpleRNN	128	256	1	0.3	3,137,621
BiRNN	128	256 (x2)	1	0.3	5,257,045
LSTM	128	256	1	0.3	3,434,069
GRU	128	256	1	0.3	3,335,253

Note on Parameters: BiRNN possesses 5,257,045 parameters—significantly more than SimpleRNN (3.14M), LSTM (3.43M), and GRU (3.34M). This occurs because bidirectionality doubles the recurrent weights (197K vs 98K) and doubles the classifier projection dimension (512 -> 7,893), requiring over 4.04M parameters in the linear classification layer alone.

4. Quantitative Experimental Results

All models were trained using Adam (lr = 0.001), batch size 128, gradient clipping (1.0), and early stopping with patience 3 on validation loss. All timings were recorded using precise torch.cuda.synchronize() barriers on an NVIDIA GeForce RTX 4060 Laptop GPU.

Model	Parameters	Train Loss	Val Loss	Test Loss	Perplexity	Top-1 Acc	Total Time
SimpleRNN	3,137,621	3.8841	5.1839	5.2490	190.38	17.19%	35.2s
BiRNN	5,257,045	3.9122	5.2446	5.3136	203.09	16.14%	30.1s
LSTM	3,434,069	3.6012	5.0571	5.1283	168.74	18.41%	32.0s
GRU	3,335,253	3.5652	5.0407	**5.0919**	**162.70**	**18.73%**	**29.1s**

Key Quantitative Findings:
1. GRU achieved the best overall performance, reaching the lowest test loss (5.0919), lowest test perplexity (162.70), highest top-1 accuracy (18.73%), and fastest total training time (29.1s across 6

epochs).
2. LSTM was a very close second, securing the lowest validation loss (5.0571) during epoch 4 and the highest top-5 accuracy (39.85%), with a perplexity of 168.74.
3. SimpleRNN suffered from vanishing gradient constraints, trailing gated models with a test perplexity of 190.38 and top-1 accuracy of 17.19%.
4. BiRNN exhibited the worst test perplexity (203.09) and lowest top-1 accuracy (16.14%) despite having 67% more parameters. Because autoregressive language modeling is strictly causal, the backward recurrent pass across the prompt prefix cannot provide any lookahead beyond the target, causing excess parameter redundancy and early overfitting on this corpus.

5. Text-Quality Metrics & Sampling Temperature Analysis

Autoregressive generation was conducted with 4 fixed Sherlock Holmes seed prompts for 60 tokens across three temperatures: T = 0.5, 1.0, and 1.5. Quality was evaluated via Distinct-1, Distinct-2, Repeated Trigram Ratio, and Unique Word Ratio.

Model	Temp (T)	Distinct-1	Distinct-2	Rep-Trigram Ratio	Unique-Word Ratio
SimpleRNN	0.5	0.613	0.877	0.034	0.661
SimpleRNN	1.0	0.758	0.979	0.000	0.821
SimpleRNN	1.5	0.904	1.000	0.000	0.937
BiRNN	0.5	0.575	0.818	0.086	0.614
BiRNN	1.0	0.721	0.996	0.000	0.804
BiRNN	1.5	0.925	1.000	0.000	0.955
LSTM	0.5	0.638	0.869	0.039	0.673
LSTM	1.0	0.758	0.983	0.000	0.820
LSTM	1.5	0.892	1.000	0.000	0.928
GRU	0.5	0.588	0.877	0.039	0.606
GRU	1.0	0.746	0.979	0.000	0.806
GRU	1.5	0.913	1.000	0.000	0.954

Temperature Effects:
• Low Temperature (T = 0.5): Sharp probability peaks favor high-frequency function words ('the', 'have', 'been', 'matter'). Syntax is relatively stable, but repeated-trigram ratio reaches 0.034–0.086 due to periodic loops ('I have no doubt that I have only no doubt').

• Moderate Temperature (T = 1.0): Optimal operating point. Zero repeated trigrams, distinct-1 rises to 0.72–0.76, and the models generate authentic Sherlock Holmes vocabulary ('Tottenham Court Road', 'Baker Street', 'the Speckled Band').
• High Temperature (T = 1.5): Flattens probability distribution across all 7,893 tokens. Lexical diversity peaks (Distinct-1 > 0.90, Distinct-2 = 1.0), but grammatical syntax degrades into disjointed word sequences.

6. Qualitative Evaluation & Coherence Scoring

Based strictly on inspection of the generated text samples, each model is evaluated on a 1–5 Coherence Scale:
• SimpleRNN (Score: 2.5/5): Generates short grammatical clauses ('I have seen my friend and now? Asked Holmes.') but quickly loses thematic consistency. Suffers from repetitive stutters at T=0.5 and severe semantic degradation at T=1.5.
• BiRNN (Score: 2.0/5): Displays the highest repetition traps at T=0.5 (repeated trigram ratio: 0.086). Its backward pass provides no causal advantage in next-word generation, frequently looping on phrases like 'at the house, and the commissionaire in the house.'
• LSTM (Score: 3.8/5): Demonstrates superior long-range dependency tracking and consistent Victorian dialogue syntax ('You will find it on the matter? He asked. It is a small thing, and we shall see nothing of the matter. I had no doubt that I should be able to do so.'). Maintains subject-verb agreement across multiple clauses.
• GRU (Score: 4.0/5): Produces the most natural sentence rhythm, idiomatic phrasing, and narrative cohesion at T=1.0 ('You hear your best? Yes. Then you wish it? Oh? He didn't being for his chair to fear else: his account it was also instructive? Asked Holmes.'). Balances fluency and thematic vocabulary effectively.

7. Conclusion & Model Recommendation

Winning Model: GRU (Gated Recurrent Unit)
Justification: GRU achieves the optimal balance of quantitative rigor and qualitative fluency. It attained the lowest test loss (5.0919), lowest perplexity (162.70), highest top-1 accuracy (18.73%), fastest training speed (29.1s), and highest qualitative coherence score (4.0/5). Its streamlined gating mechanism (reset and update gates) mitigates vanishing gradients while avoiding the parameter inflation of LSTM and BiRNN.

Project Limitations:
1. Corpus Scale: 121,193 tokens is small for word-level language modeling, limiting deep syntactic generalization.
2. Fixed Vocabulary: The ~8,000 token vocabulary cannot generate out-of-vocabulary named entities or morphological variations.
3. Recurrent Bottleneck: Sequential hidden-state compression limits memory span compared to modern Transformer self-attention.